HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and simulated wide-bandwidth RF low-noise amplifiers and Analog-to-Digital converter blocks using **Cadence** tools for bio-sensing applications, meeting stringent noise figure requirements.
- Developed AI assisted design methods to automate the simulation, design, and layout of analogue circuits, integrating custom and industry-standard Electronic Design Automation (EDA) tools.
- Collaborated with experienced layout engineers to implement analogue circuit layouts, ensuring optimal matching, shielding, and grounding for integrated circuits.
- Prepared technical presentations and reports detailing circuit design methodologies, simulation results, and project progress for senior Tyndall design team members.
- Contributed to project management by tracking task completion and providing regular updates on design milestones and potential challenges.
- Wrote Python scripts to automate data extraction and preliminary analysis from simulation output files, reducing manual processing time by approximately **30%**.
- Containerised circuit simulation environments using **Docker** to ensure consistent and reproducible testing conditions across different development machines.

SKILLS

Circuit Design – RF circuits, Analogue circuits, Mixed-Signal circuits, Amplifiers, Low Noise Amplifiers, Analog-to-Digital Converters, Schematic Design, Simulation, PVT Corners

EDA Tools – Cadence, Siemens-Mentor, Synopsys

AI/ML – AI assisted design, Machine Learning Algorithms

Programming – Python, Shell Scripting

Software & Tools – Docker, Git, Jira, Confluence, Linux

General – Integrated Circuit Layout, Technical Documentation, Report Writing, Presentation Skills

PROJECTS

AI-Assisted RF Amplifier Design

- Developed an AI-driven framework to explore design spaces for RF low-noise amplifiers, identifying optimal component configurations that improved signal-to-noise ratio by an average of **15%**.
- Integrated **Python** scripts with **Cadence** simulation environments to automate parameter sweeps and performance analysis for custom amplifier designs.
- Documented the AI model architecture and integration process, creating a reusable component for future analogue circuit design projects.

Bio-Signal ADC Block Design

- Designed and simulated an Analog-to-Digital Converter (ADC) block using **Siemens-Mentor Graphics** tools, targeting capture of bio-electrical signals like ECG and EMG.
- Performed detailed analysis across Process, Voltage, and Temperature (PVT) variations to ensure design robustness for the bio-sensing application.
- Generated layout specifications and collaborated with senior engineers to ensure efficient and functional physical implementation of the ADC block.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)